

Digital communication

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Book chapters

Forouzan: 4.1, 5.1

Kihl: 3.2, 3.4

Data transfer on a link

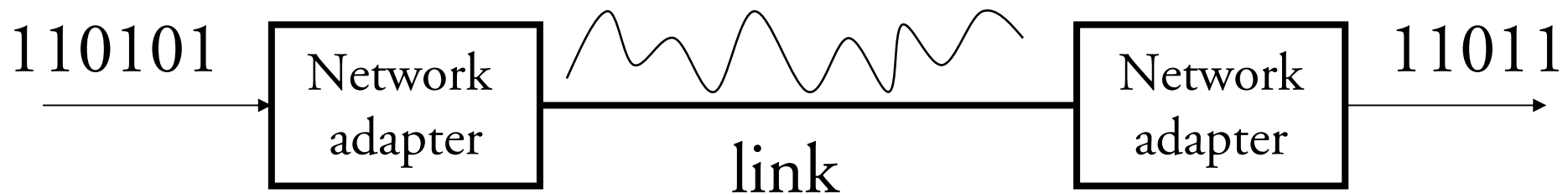


How can we represent the bits as signals when they are transmitted on a link?

Sender and receiver must use the same rules for transmission.

This is called the **Physical Layer protocol**.

Digital communication



- Digital transmission: The bits are represented by digital signals.
- Analog transmission: The bits are represented by analog signals.

Transmission of digital data



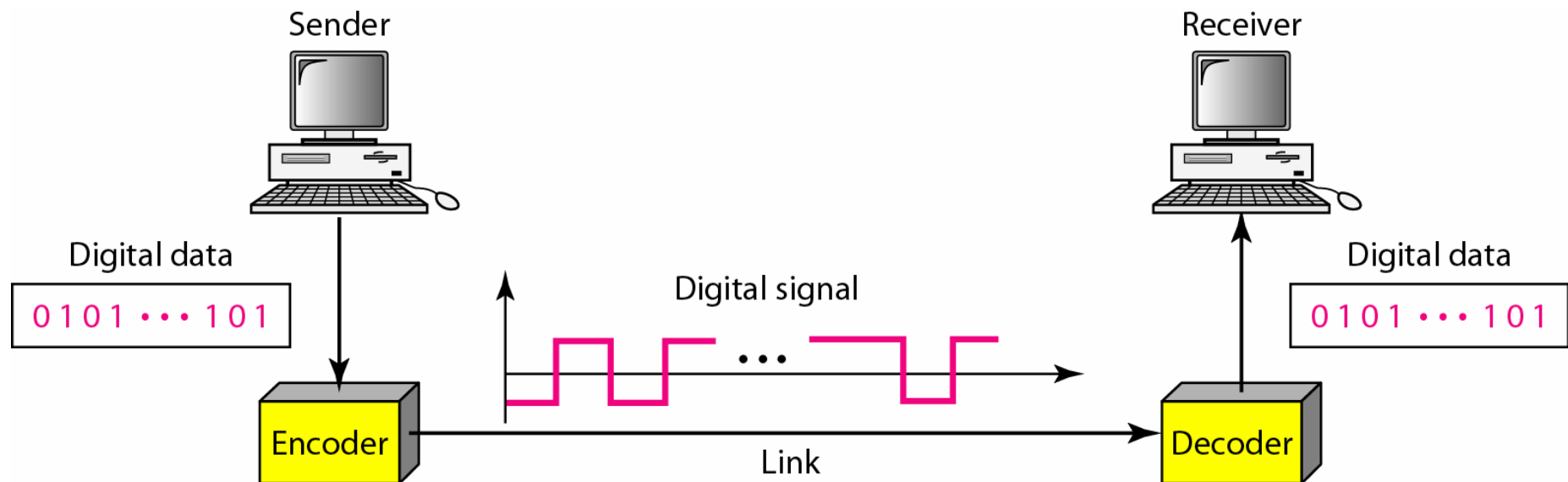
All data transfer occur on a *channel* with a certain bandwidth ($f_2 - f_1$).

Baseband transmission: $f_1 = 0$

Broadband transmission: $f_1 > 0$

Digital transmission

Line coding is the process of converting digital data to digital signals. The bits are represented by different voltage amplitudes.



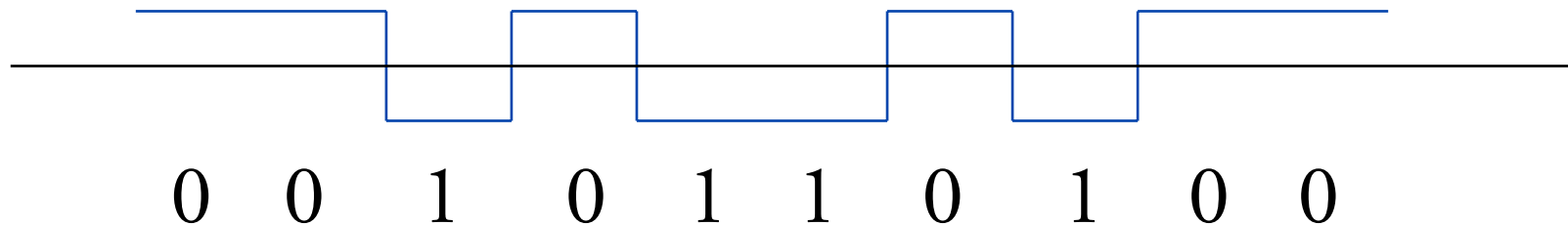
Three line coding schemes

- Non-Return-to-Zero Level (NRZ-L)
- Manchester
- Differential Manchester

Non-return to zero (NRZ-L)

0 = high voltage amplitude

1 = low voltage amplitude

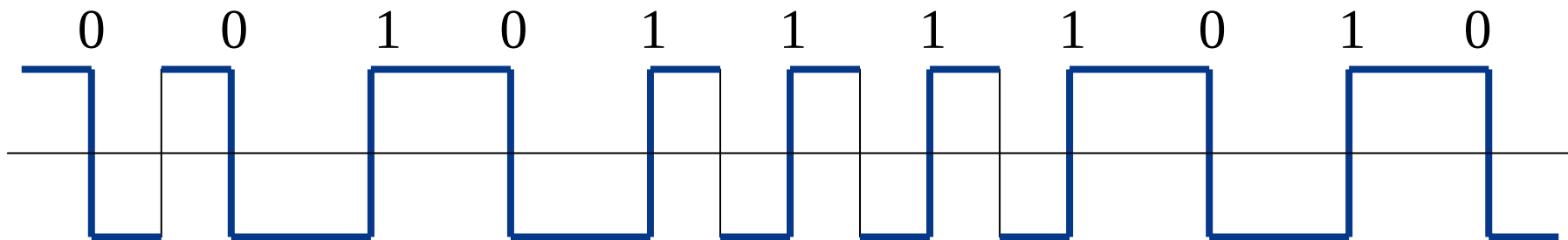


Problems with synchronization!

Manchester

Combines NRZ with a clock pulse.

No problems with synchronization.

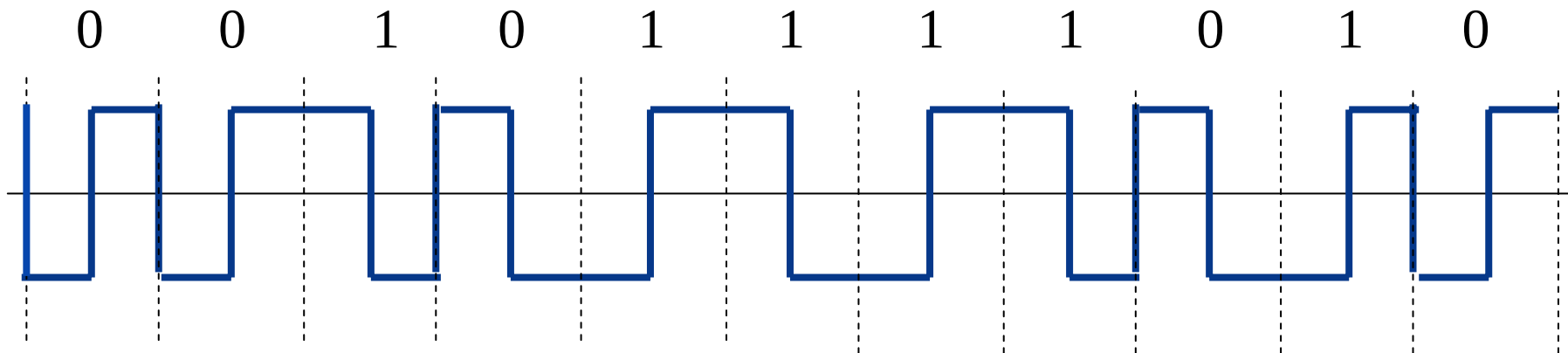


Signal rate is twice as high as in NRZ.

Differential Manchester

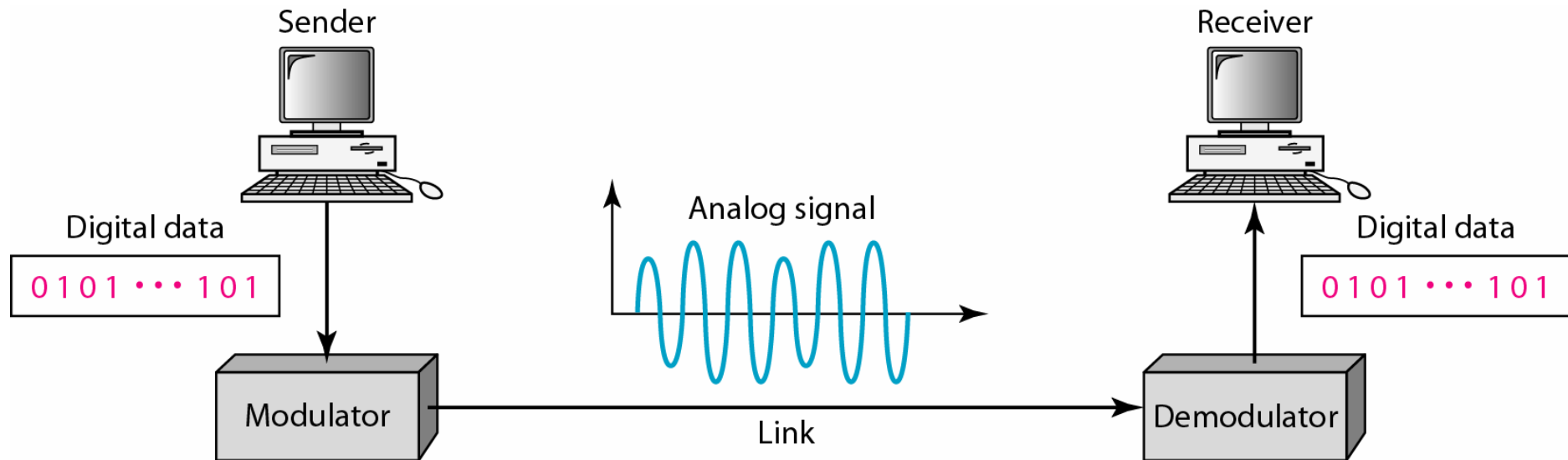
0 = Inversion in the beginning of the bit

1 = No inversion in the beginning of the bit

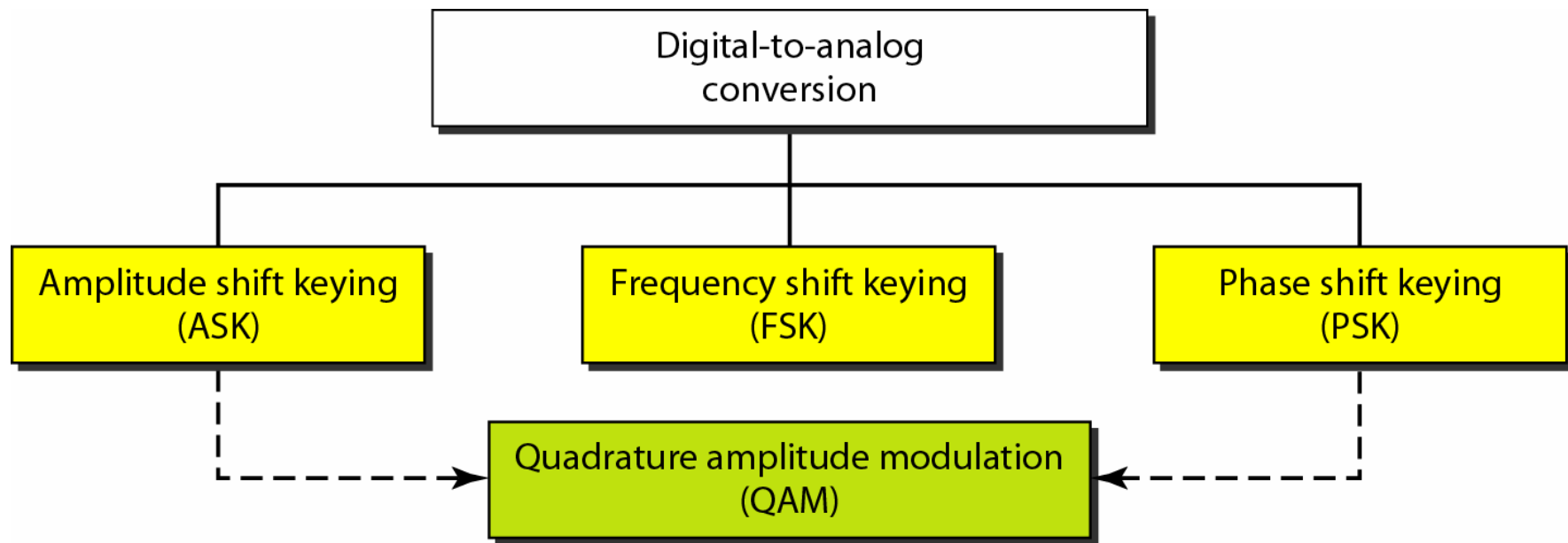


Analog transmission

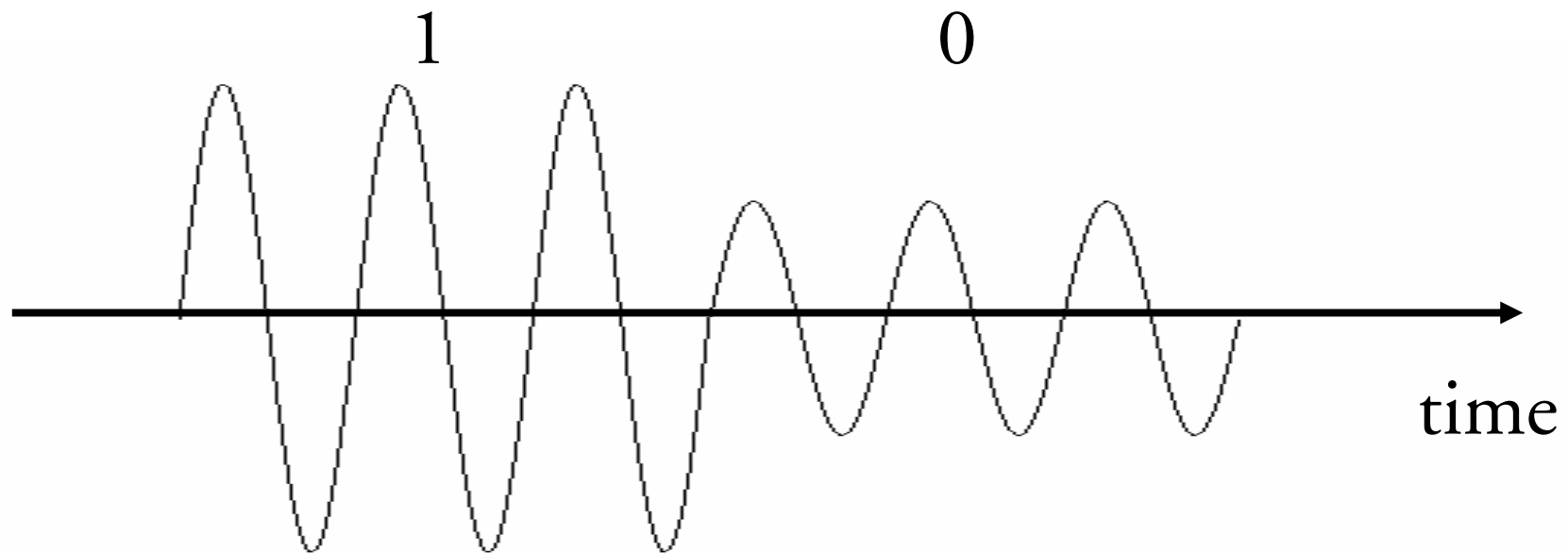
Analog transmission uses **modulation**. The digital data is represented by sinus waves. The signal is modulated on a **carrier signal** (swe: bärvåg).



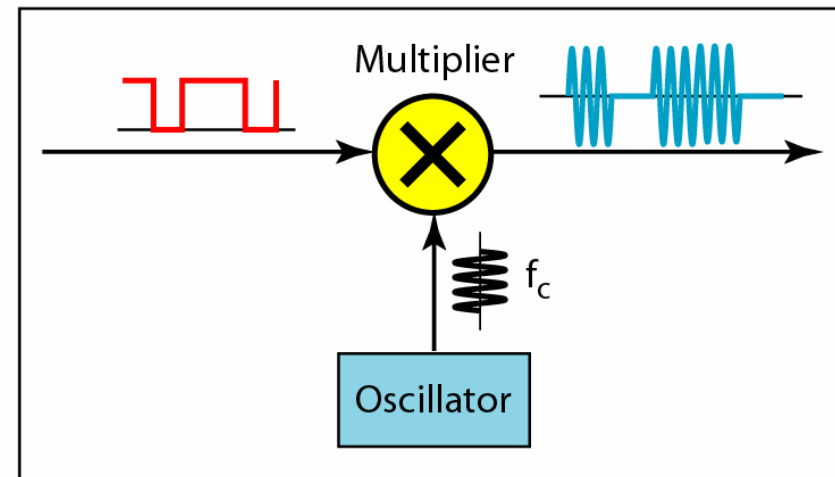
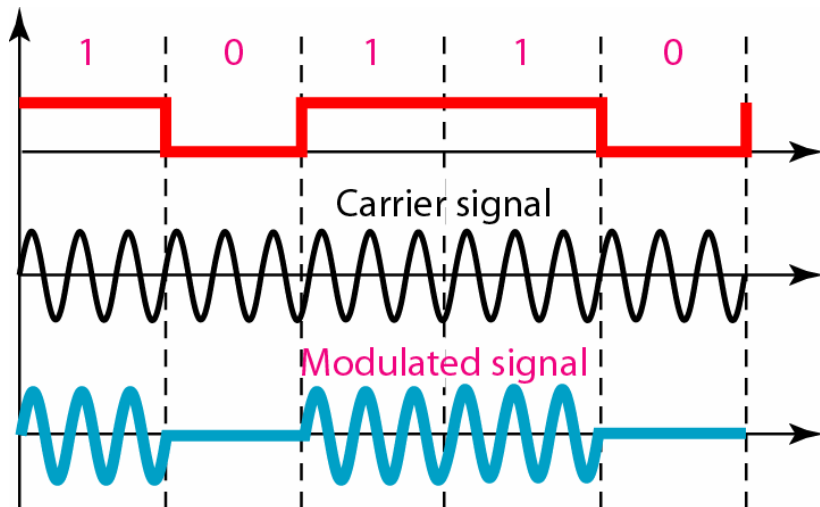
Types of modulation schemes



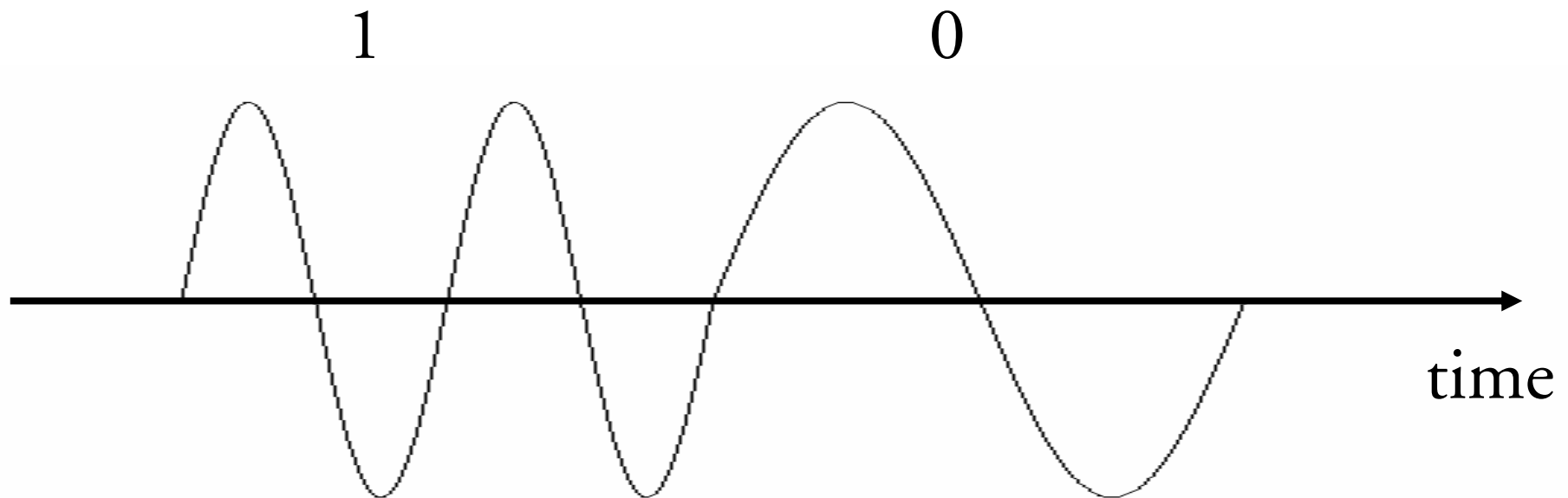
Amplitude Shift Keying (ASK)



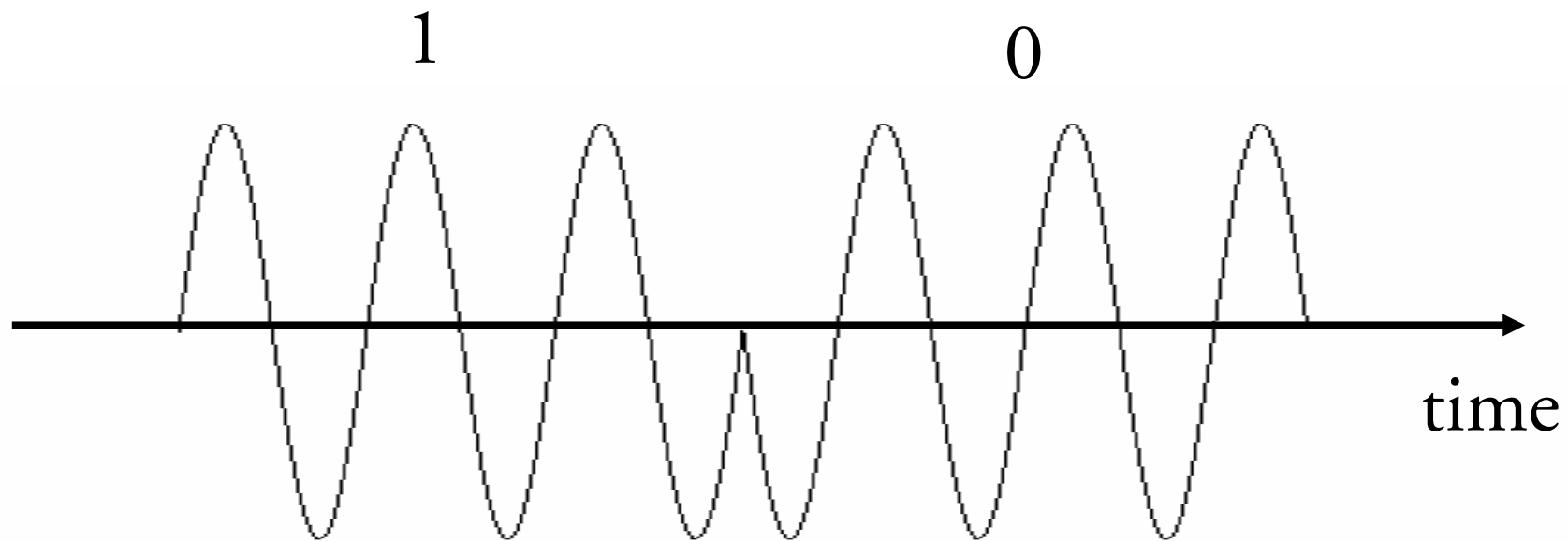
Modulation process for ASK



Frequency Shift Keying (FSK)



Phase Shift Keying (PSK)



Quadrature Phase Shift Keying (QPSK)

